Machine Learning / Deep Learning Project Ideas in FinTech

Indian Market Focus

Overview

This document presents **10 well-defined Machine Learning (ML) and Deep Learning (DL) project ideas in the FinTech domain**, specifically tailored to **gaps in the Indian financial ecosystem**.

Each idea includes:

- Description
- Identified Market Problem
- Literature Survey
- Conclusion (Value Proposition)
- Simplified Technical Implementation

These projects are suitable for **final-year academic work, research publications, startup validation, and investor pitches**.

1. Alternative Credit Scoring for Thin-File Customers Using Mobile & Behavioral Data

Description

This project develops a machine learning–based credit scoring system for individuals lacking formal credit history ("thin-file" users). The model leverages **mobile phone usage patterns, recharge behavior, app activity, and limited transaction data** to predict repayment probability.

Market Problem

A large portion of India's population remains underbanked due to reliance on bureau-based credit scoring systems, resulting in:

- Credit rejections
- Overpriced loans
- Dependence on informal lenders

Literature Survey

- Björkegren et al., *Behavior Revealed in Mobile Phone Usage Predicts Loan Repayment*
- ACM research on alternative data for credit scoring
- Surveys on big-data-driven financial inclusion

Conclusion

This solution promotes **financial inclusion**, improves approval rates, and reduces default risk. It is commercially viable, policy-aligned, and academically publishable.

Simplified Implementation

1. Consent-based mobile metadata and repayment labels
2. Feature aggregation and anonymization
3. Models: XGBoost, LSTM/Transformer, optional graph features
4. Metrics: AUC, calibration, expected loss
5. Explainability via SHAP
6. Deployment through REST APIs with drift monitoring

2. Real-Time UPI Fraud Detection Using Graph Neural Networks

Description

A graph-based fraud detection framework where accounts are nodes and transactions are edges, using **temporal Graph Neural Networks** to detect fraud rings in real time.

Market Problem

- Rule-based systems fail against coordinated fraud
- Rapid evolution of UPI scam patterns

Literature Survey

- Graph Neural Networks for financial fraud
- Temporal Graph Networks (TGN)
- ASA-GNN frameworks

Conclusion

Graph-based deep learning provides superior fraud detection with lower false positives.

Simplified Implementation

1. Transaction stream ingestion
2. Sliding window graph construction
3. GraphSAGE / GAT with temporal attention
4. Metrics: Precision@K, false-positive rate
5. Low-latency inference pipeline

3. AML & Suspicious Transaction Detection Using Unsupervised Deep Learning

Description

An unsupervised AML monitoring system using autoencoders and anomaly detection to flag suspicious behavior without heavy reliance on labeled SAR data.

Market Problem

- High false alert volumes
- Scarcity of labeled AML data
- Rapid laundering strategy evolution

Literature Survey

- Autoencoder-based anomaly detection
- AI-driven AML system surveys

Conclusion

Unsupervised learning enables discovery of novel laundering patterns and reduces investigation load.

Simplified Implementation

1. Transaction aggregates and network features
2. Behavioral normalization
3. LSTM/Transformer autoencoders, Isolation Forest
4. Metrics: SAR recall, alert reduction
5. Batch scoring with explainability dashboards

4. MSME Invoice Financing Risk Assessment Using ML & NLP

Description

A system to validate invoice authenticity, predict payment delays, and score invoices for financing using **OCR, NLP, GST verification, and graph analysis**.

Market Problem

Manual invoice verification and high perceived risk restrict MSME access to working capital.

Literature Survey

- Invoice payment prediction models
- MSME credit gap research

Conclusion

Unlocks scalable working capital for MSMEs and reduces lender risk.

Simplified Implementation

1. Invoice PDFs and GST data ingestion
2. OCR and structured extraction
3. CNN + NLP for validation, XGBoost for delay prediction
4. Metrics: Parsing accuracy, RMSE
5. API-based lender integration

5. Early Warning System for Loan Delinquency in NBFCs

Description

Predict loan stress 30–90 days in advance using borrower behavioral and transaction signals.

Market Problem

Delinquencies are detected too late, leading to reactive collections.

Literature Survey

- Early warning indicators in credit risk
- ML-based loan monitoring systems

Conclusion

Early intervention reduces NPAs and improves portfolio health.

Simplified Implementation

1. Loan ledgers and transaction streams
2. Rolling behavioral indicators
3. Survival analysis and LSTM/Transformer models
4. Metrics: Lead-time gain
5. CRM integration

6. Reinforcement Learning–Based Robo Advisor for SIP Optimization

Description

A goal-driven robo-advisor using reinforcement learning to dynamically optimize SIP allocations considering risk and tax constraints.

Market Problem

Existing robo-advisors lack personalization and tax-aware optimization.

Literature Survey

- Deep RL for portfolio optimization
- Robo-advisory adoption studies

Conclusion

Combines SIP behavior with RL, making it both investable and publishable.

Simplified Implementation

1. Mutual fund NAVs and user profiles
2. PPO/DDPG agents
3. Risk-adjusted reward functions
4. Backtesting vs benchmark SIPs
5. Explainable advisory UI

7. Merchant Credit Scoring for BNPL Platforms

Description

A real-time underwriting system for merchant BNPL using sales, refunds, logistics, and KYC data.

Market Problem

Rapid BNPL growth lacks robust merchant risk models.

Conclusion

Improves BNPL capital efficiency and default control.

Simplified Implementation

1. Merchant operational data
2. Ensemble ML with anomaly detection
3. Metrics: Approval lift vs defaults
4. Scoring API deployment

8. Churn & Lifetime Value Prediction for Digital Wallets

Description

Predict user churn and lifetime value using transaction and engagement signals.

Market Problem

High customer acquisition cost and poor retention targeting.

Conclusion

Improves unit economics and campaign ROI.

Simplified Implementation

1. App sessions and transaction logs
2. XGBoost + Transformer models
3. Lift-based evaluation
4. CRM-integrated recommendations

9. Predictive Credit Lines for E-Commerce Sellers

Description

Dynamic working-capital credit limits using probabilistic sales forecasting.

Market Problem

Static limits ignore demand volatility and seasonality.

Conclusion

Balances seller liquidity with lender risk.

Simplified Implementation

1. Sales and inventory data
2. DeepAR / Transformer forecasting
3. Risk-constrained optimization
4. Daily credit limit updates

10. Multimodal Crop Insurance Claim Verification Using Satellite Imagery

Description

Automated crop insurance claim verification using satellite imagery, weather data, and claim metadata.

Market Problem

Manual verification is slow, costly, and error-prone.

Conclusion

Reduces fraud and accelerates claim settlement.

Simplified Implementation

1. Sentinel/Landsat imagery and claims data
2. NDVI time-series extraction
3. CNN/Transformer with tabular fusion
4. Fraud detection accuracy metrics
5. Claim triage dashboard